Unit : 3 PROGRAMS

- Write a JSP to print Hello World.

```
<%@ page language="java" contentType="text/html; charset=UTF-8"
pageEncoding="UTF-8"%> <!DOCTYPE html>
<html>
<head> <title>Hello World</title> </head>
<body>
<h1>Hello World</h1>
 </body>
 </html>
```

- Write a JSP that accepts a year from the user and print whether it is a leap year or not.

```
<%@ page language="java" contentType="text/html; charset=UTF-8"
   pageEncoding="UTF-8"%>
<!DOCTYPE html>
<html>
<head></head>
<body>
   <h2>Leap Year Checker</h2>
   <form method="post">
      Enter Year:
      <input type="number" name="year" required>
      <input type="submit" value="Check">
   </form>
   <%
      String yearStr = request.getParameter("year");
      if (yearStr != null) {
         int year = Integer.parseInt(yearStr);
```

```
            if ((year % 400 == 0) || (year % 4 == 0 && year % 100 != 0)) {
        %>
            <h3><%= year %> is a Leap Year.</h3>
        <%
            } else {
        %>
            <h3><%= year %> is Not a Leap Year.</h3>
        <%
            }
        }
        %>
    </body>
    </html>
```

- Write a JSP that performs page directive.

```
<%@ page language="java" %>
<html>
<body>
    <h2>Page Directive Example</h2>
    <% out.println("Hello World!"); %>
</body>
</html>
```

- Write a JSP that performs include directive.

First.jsp

```
<html>
<head></head>
<body>
Username : <input type="text" name="txtnm" value="">
Password : <input type="text" name="txtps" value="">
<input type="submit" name="submit" value="submit">
</body>
</html>
```

```
<%@include file="First.html" %>

<html>

<head> </head>

<body>

<h1>welcome to include directive example</h1>

</body>

 </html>
```

- Write a JSP that performs all the Scripting Elements.

```
<%@ page language="java" contentType="text/html; charset=UTF-8"
pageEncoding="UTF-8"%>

<html>

<head> </head>

<body>

<h2>JSP Scripting Elements</h2>

<%!

        // 1. Declaration

        String name = "Hello";

        int square(int n)

         { return n * n; }

 %>

<%

        // 2. Scriptlet

        int number = 5;

         int result = square(number);

%>

<!-- 3. Expression -->
```

```
<p>Name: <%= name %></p>

<p>Number: <%= number %></p>

<p>Square: <%= result %></p>

</body>

</html>
```

- Write a JSP programs and create javaBean

Student.java

```
package mypackage;
public class Student {
        private String name;
        public Student() {
        }
        public void setName(String name)
        { this.name = name; }
         public String getName()
         { return name;
        }
}
```

student.jsp

```
<jsp:useBean id="student"
                class="mypackage.Student"
                scope="request" />
<jsp:setProperty name="student"
                    property="name"
                    value="Hello" />
<html>
        <body>
        <h2>Student Information</h2>
        Student Name:
```

```
        <jsp:getProperty name="student" property="name" />
        </body>
    </html>
```

- Write a JSP Expression Language Programs.

```
<html>
    <body>
        <h2>EL Arithmetic Operations</h2>
        <p>Addition: ${10 + 5}</p>
        <p>Subtraction: ${10 - 5}</p>
        <p>Multiplication: ${10 * 5}</p>
        <p>Division: ${10 / 5}</p>
    </body>
</html>
```

JSTL

- First Add the dependency for JSTL.

```
<dependency>
  <groupId>jakarta.servlet.jsp.jstl</groupId>
  <artifactId>jakarta.servlet.jsp.jstl-api</artifactId>
  <version>3.0.1</version>
</dependency>

<dependency>
  <groupId>org.glassfish.web</groupId>
  <artifactId>jakarta.servlet.jsp.jstl</artifactId>
  <version>3.0.1</version>
</dependency>
```

- Simple JSTL Program.

```
<%@ page language="java"
  contentType="text/html; charset=UTF-8"
  pageEncoding="UTF-8"%>
```

```
<%@ taglib prefix="c" uri="jakarta.tags.core" %>
<html>
<head></head>
<body>
   <h2>JSTL Example</h2>
   <!-- Set a variable -->
   <c:set var="name" value="Yash"/>

   <!-- Display the variable -->
   <p>Name: <c:out value="${name}"/></p>

   <!-- Conditional statement -->
   <c:if test="${10 > 5}">
      <p>10 is greater than 5</p>
   </c:if>
</body>
</html>
```

- Write a program of JSTL forEach.

```
<%@ page language="java"
   contentType="text/html; charset=UTF-8"
   pageEncoding="UTF-8"%>

<%@ taglib prefix="c" uri="jakarta.tags.core" %>
<html>
<body>

   <h2>JSTL Loop Example</h2>
   <c:forEach var="i" begin="1" end="5">
```

```
        <p>Number: <c:out value="${i}"/></p>

      </c:forEach>

    </body>
    </html>
```

- Write a program of JSTL c:forTokens.

```
    <%@ page language="java"
      contentType="text/html; charset=UTF-8"
      pageEncoding="UTF-8"%>

    <%@ taglib prefix="c" uri="jakarta.tags.core" %>

    <html>
    <head></head>
    <body>
      <h2>JSTL c:forTokens Example</h2>
      <c:set var="subjects" value="Java,Python,HTML,JavaScript,SQL"/>
      <p>Subjects:</p>
      <c:forTokens items="${subjects}" delims="," var="subject">
        <p>
          <c:out value="${subject}"/>
        </p>
      </c:forTokens>
    </body>
    </html>
```

- Write a program of JSTL c:choose, c:set, c:remove.

```
    <%@ page language="java"

      contentType="text/html; charset=UTF-8"

      pageEncoding="UTF-8"%>

    <%@ taglib prefix="c" uri="jakarta.tags.core" %>

    <html>

    <body>
```

```
<c:set var="marks" value="75"/>
<c:choose>
  <c:when test="${marks >= 70}">
    <p>Grade: A</p>
  </c:when>

  <c:when test="${marks >= 50}">
    <p>Grade: B</p>
  </c:when>

  <c:otherwise>
    <p>Grade: C</p>
  </c:otherwise>
</c:choose>
<c:remove var="name"/>
</body>
</html>
```

- Write a program of JSTL c:import.

  Import.jsp

```
<%@ page language="java"
  contentType="text/html; charset=UTF-8"
  pageEncoding="UTF-8"%>
<%@ taglib prefix="c" uri="jakarta.tags.core" %>
<html>
<head></head>
<body>

  <h2>JSTL c:import Example</h2>
```

```
    <p>This is the main JSP page.</p>
    <h3>Imported Content:</h3>

    <c:import url="header.jsp"/>

        //<c:redirect url="header.jsp"/>
</body>
</html>
```

header.jsp

```
<h3>Welcome to My Website</h3>
<p>This content is imported from header.jsp</p>
```

- Write a program of JSTL c:catch.

```
<%@ page language="java" contentType="text/html; charset=UTF-8"
pageEncoding="UTF-8"%>

 <%@ taglib prefix="c" uri="jakarta.tags.core" %>

<html>

<head> </head>

<body>

<h2>JSTL c:catch Example</h2>

<c:catch var="error">

<%      int a = 10;

         int b = 0;

         int result = a / b;

         out.println("Result = " + result);

%>

        </c:catch>

        <c:if test="${not empty error}">
```

```
<p><b>Exception Occurred!</b></p>

<p> Error Message: <c:out value="${error.message}"/> </p>

</c:if>

</body>

</html>
```